



Locating and quantifying gas emission sources using remotely obtained concentration data

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Outline

Motivation

- A method for detecting, locating and quantifying sources of gas emissions to the atmosphere
- From remotely obtained atmospheric gas concentration measurements

Issues

- Potentially **large** background gas concentrations ($\approx 1800ppb$ for CH_4)
- Need to detect small signals ($\approx 5 - 35ppb$ for CH_4)
- Gas dispersion determined by prevailing wind conditions

Approach

- Plume model represents gas dispersion between source and measurement location
- Measured concentration is sum of contributions from sources and **relatively smooth** background
- Infer source locations, source emission rates, background level, plume biases and uncertainties



Survey aircraft ($\approx 50\text{ms}^{-1}$, $\approx 200\text{m}$ above ground)

Motivating test applications

Synthetic problem

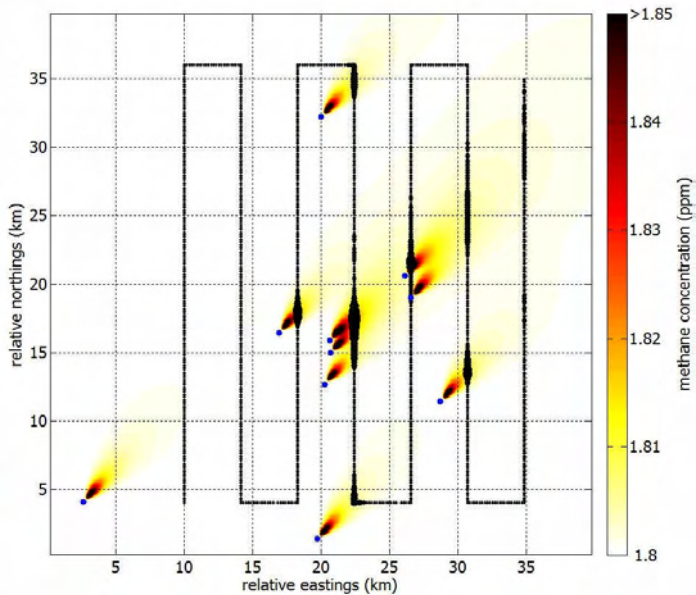
- **Known** wind field, sources and background, 10 sources

Landfill

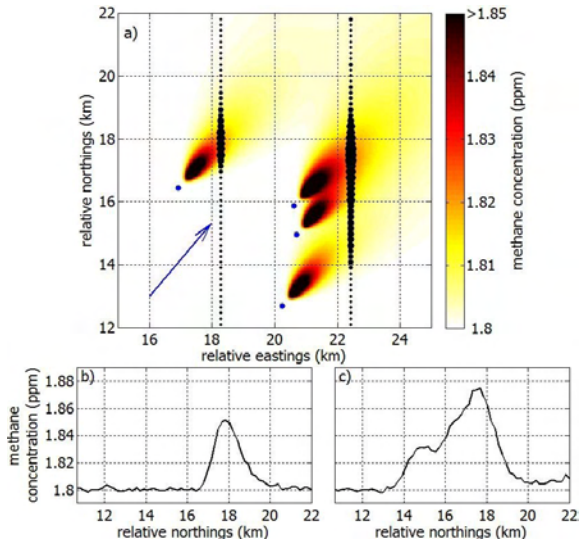
- 2 landfill regions, probable diffuse sources
- Wind field from UK met-office **global circulation model**

Flare stack

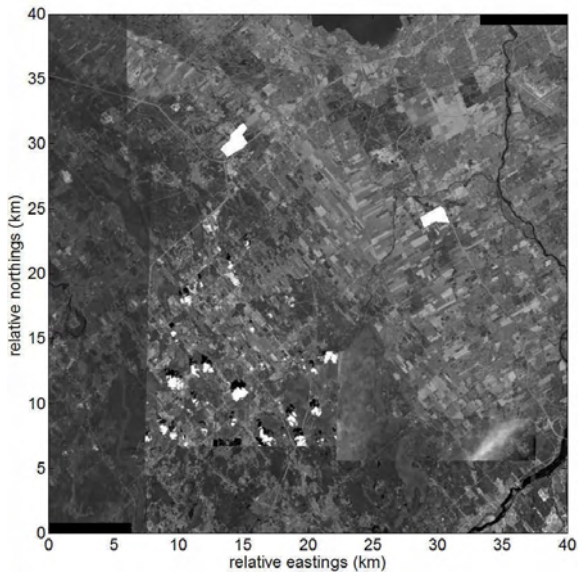
- Single elevated near-point source
- Wind field from UK met-office global circulation model
- **Coastal** location



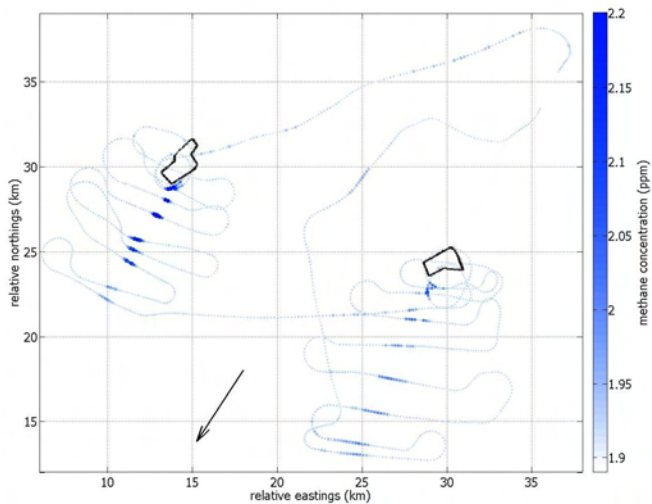
Synthetic problem revealed



(a) two passes x-y (b) first pass in time (c) second pass in time



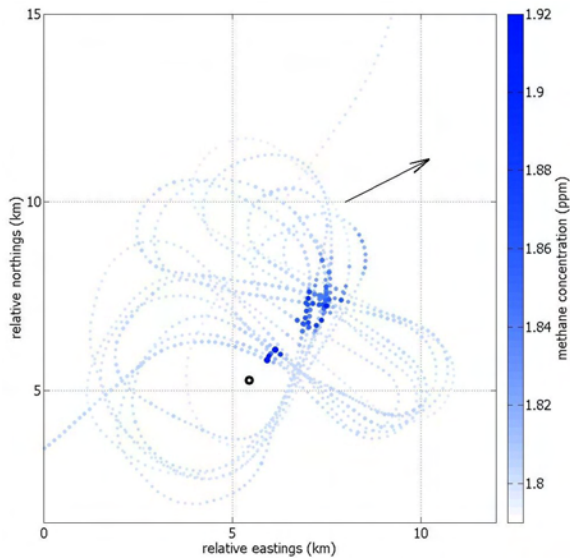
Landfill from above



Landfill measurements



Flare stack



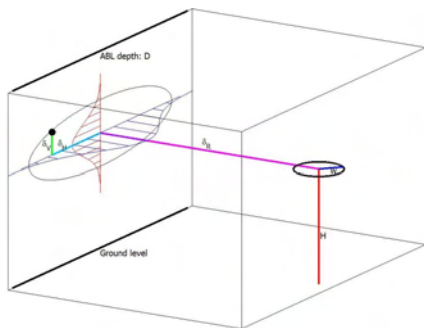
Flare stack measurements (wind direction bias)

Model formulation

$$\mathbf{y} = \mathbf{A}\mathbf{s} + \mathbf{b} + \epsilon$$

- $\mathbf{y}$: measured concentrations
- $\mathbf{A}$: assumed known from plume model
- $\mathbf{s}$: sources to be estimated
- $\mathbf{b}$: background to be estimated
- ϵ : measurement error (assumed Gaussian), variance to be estimated

Plume model



- Red: Source height H
- Blue: Source half-width w
- Magenta: Downwind offset δ_R
- Cyan: Horizontal offset δ_H
- Green: Vertical offset $\delta_V = z - H$
- **ABL** height: D
- Horizontal extent: $\sigma_H = \delta_R \tan(\gamma_H) + w$
- Vertical extent: $\sigma_V = \delta_R \tan(\gamma_V)$
- *Opening angles:* γ_H, γ_V

$$a = \frac{1}{2\pi|\mathbf{U}|\sigma_H\sigma_V} \exp\left\{-\frac{\delta_H^2}{2\sigma_H^2}\right\} \times \left\{ \exp\left\{-\frac{(z-H)^2}{2\sigma_V^2}\right\} + \exp\left\{-\frac{(z+H)^2}{2\sigma_V^2}\right\} \right. \\ \left. + \exp\left\{-\frac{(\mathbf{2D} - z - H)^2}{2\sigma_V^2}\right\} + \exp\left\{-\frac{(2D - z + H)^2}{2\sigma_V^2}\right\} \right\}$$

Background model

Requirements

- Positive and smoothly-varying, spatially and temporally
- Basis function representation: $\mathbf{b} = \mathbf{P}\boldsymbol{\beta}$
- We use Gaussian Markov random field
- Explicit spatial dependence due to wind transport incorporated

Random field prior

$$f(\boldsymbol{\beta}) \propto \exp\left\{-\frac{\mu}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^T \mathbf{J}_{\boldsymbol{\beta}}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)\right\}$$

- $\mathbf{J}_{\boldsymbol{\beta}}$ is sparse, $\mathbf{P} = \mathbf{I}$
- Fast estimation

Inference strategy

Initial point estimation

- Sources and background
- Source locations assumed on **fixed grid**
- **Fast** estimation of starting solution for Bayesian inference

Subsequent Bayesian inference

- Sources, background, measurement error, wind-field parameters, ...
- **Grid-free** sources modelled using Gaussian **mixture model**
- Reversible jump MCMC inference
- Quantified parameter **uncertainties** and **dependencies**

Initial point estimation

Background prior

$$f(\beta) \propto \exp\left\{-\frac{\mu}{2}(\beta - \beta_0)^T \mathbf{J}_\beta(\beta - \beta_0)\right\}$$

Source prior (**Laplace**)

$$f(\mathbf{s}) \propto \exp\{-\lambda \|\mathbf{Q}\mathbf{s}\|_1\}$$

Likelihood

$$f(\mathbf{y}|\mathbf{s}, \beta) \propto \exp\left\{-\frac{1}{2\sigma_\epsilon^2} \|\mathbf{A}\mathbf{s} + \mathbf{P}\beta - \mathbf{y}\|^2\right\},$$

Posterior

$$f(\mathbf{s}, \beta|\mathbf{y}) \propto f(\mathbf{y}|\mathbf{s}, \beta)f(\mathbf{s})f(\beta)$$

Maximum a-posteriori estimate

$$\operatorname{argmin}_{\mathbf{s}, \beta} \quad \frac{1}{2\sigma_\epsilon^2} \|\mathbf{A}\mathbf{s} + \mathbf{P}\beta - \mathbf{y}\|^2 + \frac{\mu}{2}(\beta - \beta_0)^T \mathbf{J}(\beta - \beta_0) + \lambda \|\mathbf{Q}\mathbf{s}\|_1$$

Bayesian inference

Parameters

- Source locations $\mathbf{z}$, “widths” $\mathbf{w}$ and emission rates $\mathbf{s}$ for mixture of m sources
- Random field background parameters β
- Measurement error standard deviation σ_ϵ
- Wind–direction correction δ_ϕ
- Others (e.g. plume opening angles)
- Call these θ which can be partitioned $\{\theta_{\kappa}, \theta_{\overline{\kappa}}\}$

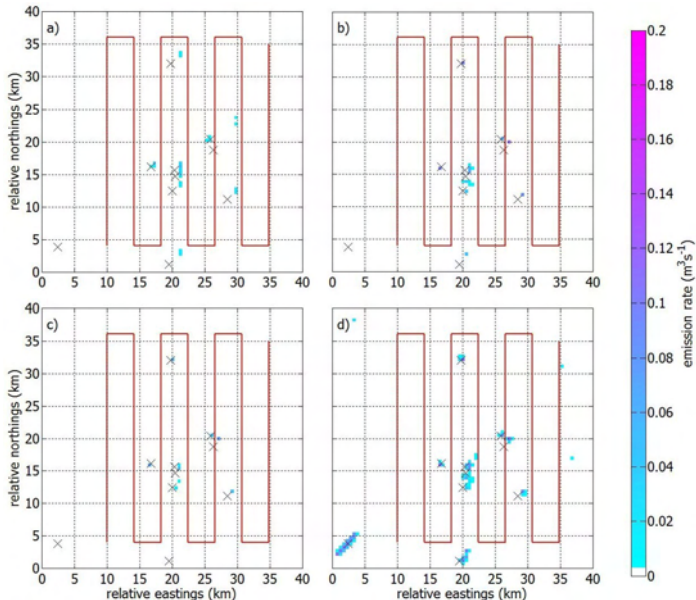
Full conditional

$$f(\theta_{\kappa} | \mathbf{y}, \theta_{\overline{\kappa}}) \propto f(\mathbf{y} | \theta_{\kappa}, \theta_{\overline{\kappa}}) f(\theta_{\kappa} | \theta_{\overline{\kappa}})$$

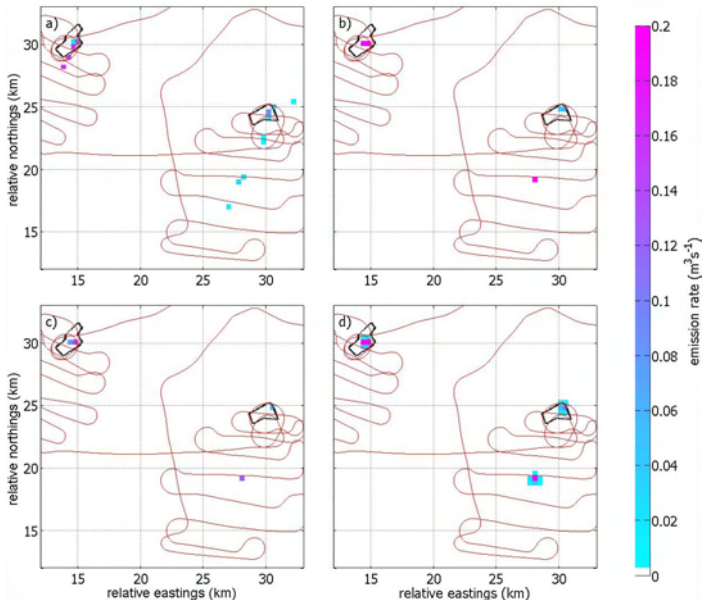
Inference tools

- Gibbs’ sampling
- **Reversible jump**
- (Metropolis–Hastings)

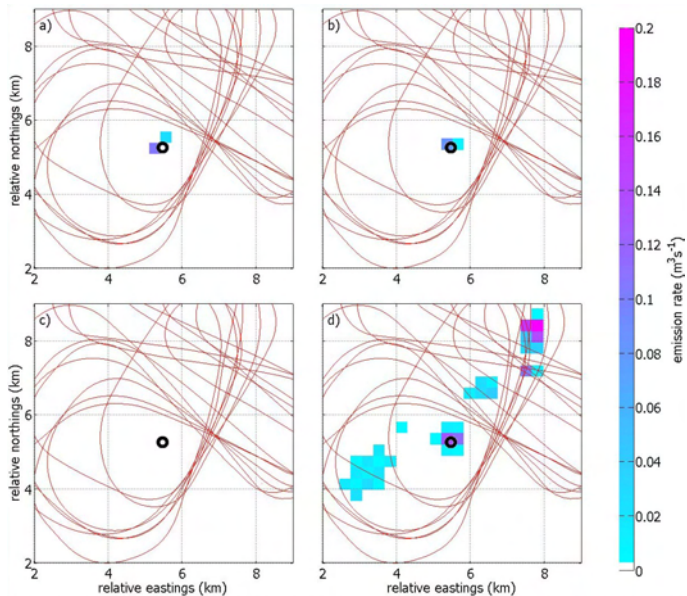
Synthetic



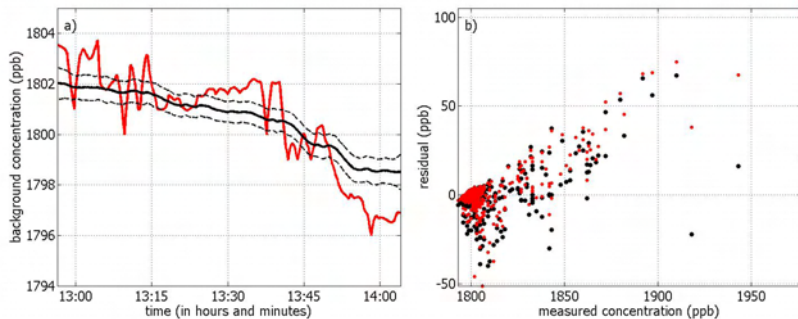
Landfill



Flare stack



Flare stack



(a) background in time (b) residual vs measured concentration
initial (red); posterior median (black)

Wind direction correction of 18°

Conclusions and on-going work

Conclusions

- **Data structure** and management
- Flexible inference using combination of standard methods
- Good performance on synthetic and field applications
- **Scalability** from iterative estimation

On-going work

- Multiple flights, multiple wind data sources
- Enhanced plume model
- Internal **calibration**
- Improved prior characterisation of sources, intermittent sources
- Simultaneous inference using multiple measurement types
- Optimal **design**
- **Line-of-sight** applications

Slides and extended abstract at www.lancs.ac.uk/~jonathan